

RAFAEL G. MADRIGAL

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Senior Machine Learning & Research Engineer

Senior ML & Research Engineer with **10 years of experience** – last 5 years in the end-to-end translation of SOTA AI Research into production systems within regulated finance. My background combines the experience as an **ML Research** investigating algorithmic feasibility with an **ML Engineer** architecting enterprise-scale solutions. This **dual perspective** enables me to lead the **full research-to-deployment lifecycle**: from the exploration of **Privacy-Enhancing Technologies and Responsible AI** for tabular ML to building hardened, modular libraries that scale **Multi-Agent Orchestration** for millions of users.

Skills

- **Generative AI & LLM Orchestration:** LangGraph, LangChain, Multi-Agent Systems, Agentic Workflows, Human-in-the-loop design, Chat Memory Management, Vector Databases (Azure AI Search, FAISS, Pinecone, Annoy)
- **Advanced Retrieval & RAG Optimization:** Query decomposition, Query Expansion, HyDE, Contextual Retrieval, OData Filtering, Custom Scoring Profiles, Hierarchical Retrieval, Lucene Queries, Semantic Caching, LLM Streaming
- **Machine Learning and Deep Learning Research:** PyTorch, TensorFlow, Transformers, SOTA LLMs (GPT-4o, Claude, Cohere), Neural Topic Modeling (TopicBERT), Recommender Systems, Supervised and Unsupervised Learning.
- **Fairness and Privacy Enhancing Technologies:** Responsible AI Frameworks (Fairlearn, IBM AIF360), Data Clean Rooms, Differential Privacy, Synthetic Data Generation, Federated Learning, AI Governance
- **Engineering & MLOps:** Python (Pandas, NumPy, SciPy, Scikit-Learn), SQL, Git, FastAPI, API Design, System Architecture, MLFlow, Arize, Evaluation Frameworks (RAGAS, Deepeval), LLM Red-teaming
- **Cloud and Data infrastructure:** Azure Open AI, AWS, Databricks, SQLMI, CosmosDB, Data Warehousing (ETL, OLAP), Power BI

WORK EXPERIENCE

Senior Machine Learning Engineer

Aug 2024 – Present

Manulife Canada

- **Cross-functionally led the architecture** 9 production Gen AI Knowledge Assistants serving 3000+ users, utilizing a hybrid architecture using LangGraph; collaborated with stakeholders to combine massive (20,000+) unstructured document collections (Azure AI Search) with large structured customer data (SQLMI) for personalized answer generation
- **Authored clean and maintainable Python core libraries** and crafted **feature-parity roadmaps** that standardized the RAG lifecycle (document pre-processing to evaluation); established **reusable code and API designs** that slashed initial UAT prototyping from 3 months to 1-2 weeks and total time-to-production from 7 months to 3 months
- **Engineered a hierarchical, two-stage retrieval architecture** (Azure AI Search) to mitigate semantic noise across large document collections (20,000+); utilized hybrid search, dynamic OData filtering, and custom Lucene queries and scoring profiles to maximize retrieval precision@k
- **Designed a few-shot query understanding pipeline leveraging LLMs** (GPT-4o) for intent classification, entity extraction, and query decomposition/ expansion; utilized inferred metadata and structural reformulations to limit the search space, drive dynamic prompting and enrich final answer quality
- **Orchestrated stateful multi-agent workflows in LangGraph** implementing **persistent memory checkpoints** (CosmosDB) to manage conversation histories and dynamic **human-in-the-loop interrupts** to autonomously resolve vague user queries
- **Optimized end-to-end system latency and reliability** by coupling CosmosDB semantic caching with GPT-4o token streaming – slashing response times for frequent queries from 30s to 200ms and reduced time-to-first-token (TTFT) from 20s to 650ms, while utilizing LLM-as-a-judge for real-time confidence scoring

Senior Machine Learning Researcher

Apr 2022 – Aug 2024

Globe Fintech Innovations (GCash)

GCash, an Ant Financial/ Alibaba company, is an all-in-one finance app (i.e., payments, wealth, insurance) with 90M+ users worldwide

- **Co-authored research accepted for an Oral Presentation at PCSC 2026:** “Defending Against Membership Inference Attacks by Contrastive Representation Learning”; produced Differentially Private synthetic data using Adaptive and Iterative Mechanism (AIM) used to reduce LiRA attack TPR using a novel joint loss function on benchmark datasets.
- **Championed the responsible use of AI** by architecting proprietary Python Library – synthesizing Fairlearn and IBM AIF360 – to standardize algorithmic bias testing, executing 30+ mathematical audits (e.g. demographic parity, equalized odds) on large-scale fraud and lending models
- Evaluated SOTA industry practices to establish the **foundational LLMOps standards** for red-teaming and model evaluation across 11 production Gen AI applications used internally, promoting **AI Safety practice** in the organization.
- Orchestrated an autonomous, **dual-node multi-agent system** (Evidence-Verifier) via **LangGraph** to automate enterprise compliance; integrated the agent with Jira via FastMCP, formulating the orchestration as a **multi-label classification** that achieved a 0.52 Macro-F1 and reduced task effort by 43% ($p < 0.05$) over an out-of-sample set of 112 project epics (880+ discrete inferences)

- Architected a **Text-to-SQL** Gen AI pipeline using LangChain and Azure OpenAI to query structured financial datasets, achieving 81% SQL generation accuracy and 87% end-to-end Q&A accuracy, securing production deployment into a consumer-facing expense tracking application
- Engineered a **Privacy Impact Assessment (PIA) Retrieval-Augmented Generation (RAG) agent** utilizing Azure OpenAI and Cohere Rerank deployed in Google Chat, achieving 86% Q&A accuracy
- Mentored 6 data scientists and 7 interns, establishing rigorous code and solution design review process to ensure the deployment of clean, maintainable ML code
- Facilitated the technical scoping and prioritization of 100+ ML use cases across 16 enterprise verticals, successfully driving the development of 11 Generative AI prototypes within a year

Data Scientist

Jul 2021 – Mar 2022

Unionbank of the Philippines

Unionbank of the Philippines took over Citi's operating business in the Philippines in 2023 and is the 9th largest bank in the Philippines

- Led the research and development of a predictive pricing POC; led 3 Data Scientists in training a LightGBM model for property valuation based on web-scraped data and OpenStreetMap geospatial features, achieving 20-58% MAPE
- Trained 6 BI analysts transitioning to Data Science in core Python programming and ML fundamentals, successfully reducing codebase rework by 50% and accelerating the delivery of internal prototypes

Early Career Experience

Apr 2016 – Jul 2021

Manila Water Philippine Ventures

- **Senior Strategy Analyst (May 2018 - July 2021):** Drove data analysis to streamline governance, reducing business case approval time by 6 weeks and saving 2,000+ executive hours. Influenced C-suite decisions in internal expansions amounting to C\$ 475M in Saudi Arabia and Myanmar
- **Business Analyst, Water Operations (Dec 2016 - May 2018):** Improved operational efficiency and customer experience through data-driven network issue resolution
- **Management Trainee (Apr 2016 - Dec 2016):** Analyzed consumer trends to enhance revenue forecasting accuracy by 10%, generating an additional C\$120k in annual revenue

PUBLICATIONS

- Agarap, A. F., Aguinaldo, M. A., **MADRIGAL, R.**, et al. (Expected April 2026). *Defending Against Membership Inference Attacks by Contrastive Representation Learning*. Full Paper & Oral Presentation, Philippine Computing Science Congress (PCSC) 2026

OTHER PROJECTS

- **Multi-Agent Assistant for AI Risk Assessments (CrewAI and Command R+).** Placer @ Cohere Build Day Hackathon (2024). Developed a multi-agent prototype with CrewAI and Cohere Command R+ to automate AI risk assessment by identifying potential ethical and data privacy concerns in project documentation
- **HF-based PEFT & Voice Cloning.** Executed parameter-efficient fine-tuning (PEFT) of Gemma-2b-it via Hugging Face; integrated voice cloning (ElevenLabs.io) for a multimodal Streamlit chat interface
- **Large-Scale Trend Analysis (182GB GDEL).** Architected a distributed pipeline using PySpark and Distributed LDA to track global geopolitical trends and topic evolution
- **Open-Source RAG Conversational Bot (Mistral & Annoy).** Engineered an end-to-end RAG application utilizing Mistral-7B-Instruct and LangChain; implemented persistent conversational memory and Annoy Vector DB for high-speed similarity search over multi-PDF uploads
- **Neural & Hierarchical Topic Modeling for Public Policy.** Dissected presidential transcripts using agglomerative and hierarchical clustering to track the longitudinal evolution of government response priorities and quantify shifts in policy focus over time

EDUCATION

Master of Science (MS) in Data Science | GPA: 3.7 on a 4.0 scale | High Distinction 2021
Asian Institute of Management

Bachelor of Science (BS) in Chemical Engineering | GPA: 3.3 on a 4.0 scale | *Cum Laude* 2015
University of the Philippines - Diliman

INTERESTS

Pursuit of Learning: enrolled in 20+ Coursera courses during the pandemic. Passionate about Personal Finance: maintained a personal transactions database for 5+ years. I play Tennis and Badminton.